MODELING AFRICAN LION MOVEMENTS AND ECOLOGICAL CORRIDORS USING MAXENT AND GRID-BASED CLUSTERING

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Abstract

African lions (**Panthera leo**), often celebrated as symbols of strength and majesty, are facing escalating threats due to habitat fragmentation driven by agriculture, urban development, and human-wildlife conflict. This study aims to model lion movement patterns and identify ecological corridors that promote safe connectivity between fragmented habitats using presence-only data and MaxEnt modeling.

We began with a dataset of 6,638 lion presence records, featuring spatial and temporal attributes. To reduce noise and computational overhead, a grid-based clustering approach was implemented using ArcGIS Pro's Fishnet tool. Several grid sizes were tested, with 10,000×5,000 yielding the best spatial representation. Post-clustering, the dataset was condensed to 1,384 representative points.

Given the nature of presence-only data, the Maximum Entropy (MaxEnt) algorithm was chosen for species distribution modeling. Environmental predictors were derived from WorldClim's bioclimatic variables, specifically: BIO16 (Precipitation of the Wettest Quarter), BIO17, BIO18, and BIO19. Climate TIFF files were converted to ASC format using QGIS for compatibility with MaxEnt.

The model achieved an AUC score of 87%, indicating strong predictive performance. Results highlighted BIO16 as the most influential predictor of lion presence, with response curves revealing bell-shaped distributions—suggesting optimal precipitation thresholds for habitat suitability. Notably, the final prediction map identified both African and ecologically similar non-African zones suitable for lion survival.

This modeling framework provides an effective blueprint for identifying and preserving ecological corridors, aiding conservation strategies that aim to reduce human-wildlife conflict and safeguard African lion populations.

Key Words: African Lions, MaxEnt, Habitat Fragmentation, Species Distribution Modeling, Ecological Corridors, Grid-based Clustering, Precipitation Variables